Automated Mitosis Detection in Histopathology

A Two-Stage Deep Learning Pipeline

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Dataset: TUPAC16

Tumor Proliferation Assessment Challenge

Motivation

Clinical Problem

- Mitosis count is key prognostic indicator
- Manual annotation is time-consuming
- High inter-observer variability
- Need for automated assessment

Technical Challenge

- Mitotic figures are small (8–128 pixels)
- Class imbalance: many non-mitotic regions
- Domain shift across staining protocols
- Requires precise localization

Dataset: TUPAC16

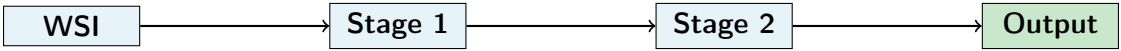
Data Characteristics

- **Source:** TUPAC16 Challenge
- **Training:** 8 slides
- **Validation:** 6 held-out slides
- **Multiple Centers:** Different scanners
- **Resolution:** 40x magnification

Preprocessing

- Macenko color normalization
- Stage 1: 64×64 patches
- Stage 2: 512×512 patches
- Total: 600 annotated mitoses
- 80% background patches excluded

Two-Stage Pipeline



Input: WSI tiles (40x)

Filter: Binary classifier on 64×64

Detect: R-CNN on 512×512

Stage 1: Binary Classifier

Architecture

- **Backbone:** EfficientNet-B3
- **Input:** 64×64 RGB patch
- **Output:** Binary probability
- **Loss:** Focal Loss

Performance

- F1 Score: **0.8894**
- Precision: 0.92
- Recall: 0.87
- Epochs: 8

Loss Function

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$$

with $\gamma = 2$, $\alpha = 0.25$

Threshold

$\tau = 0.5087$ (recall-optimized)

Flags $\sim$15% of patches

Stage 2: Object Detector

Architecture

- **Model:** Faster R-CNN
- **Backbone:** ResNet50-FPN
- **Input:** 512×512 RGB
- **Output:** Bounding boxes

Custom Anchors

- Sizes: 8, 16, 32, 64, 128 px
- For small objects
- Aspect ratios: 0.5, 1, 2

Performance

- mAP@0.5: **0.7598**
- F1 Score: **0.7742**
- Precision: 0.714
- Recall: 0.789
- Epochs: 65

Inference

- Confidence: 0.40
- NMS IoU: 0.3

Results: Single-Center Performance

Config	F1	Prec	Status
Stage 1	0.8894	0.92	Screening
Stage 2	**0.7742**	**0.714**	**BEST**
Pipeline	0.5806	0.68	Not Rec.

Key Finding

Direct detection (F1=0.7742) outperforms cascade (F1=0.5806)!
Early filtering removes legitimate mitosis cases.

Cross-Center Evaluation: Domain Shift

Slide	Center	F1
50	A	0.8571
67	B	0.8000
65	C	1.0000
66	D	1.0000
63	E	1.0000
AccentRed!30 34	F	0.2857

63% F1 Degradation

Different scanner/stain protocol on Slide 34
Critical for deployment: Multi-center validation essential

Error Analysis

False Positives (3 total)

- Overlapping nuclei
- Apoptotic bodies
- Stain artifacts
- Gallery: outputs/false_positive_gallery/

False Negatives (4 total)

- Poorly stained figures
- Telophase mitosis
- Structure overlap
- Dataset disagreement
- Gallery: outputs/false_negative_gallery/

Why Direct Detection Beats Cascade

Conventional Wisdom

Cascade reduces computation, Stage 1 filters easy negatives

Reality

Stage 1 threshold removes legitimate mitoses!
Some difficult but valid cases fall below $\tau = 0.5087$

Insight

For small object detection with class imbalance:
End-to-end learning $>$ Staged approaches
Deploy Stage 2 directly for best performance

End-to-End Pipeline

Inference Workflow

1. Load WSI at 40x magnification
2. Sliding window: 512×512 patches
3. Run Faster R-CNN (recommended)
4. Slide-level NMS (IoU=0.3)
5. Output: Bounding boxes + scores
6. Nottingham score (clinical grading)

Nottingham Grading

- Score 1: ≤ 7 mitoses per 10 HPF
- Score 2: 8–14 mitoses per 10 HPF
- Score 3: ≥ 15 mitoses per 10 HPF

Code Organization

Model Architecture

`models/`

- stage1_classifier.py
- stage2_detector.py

Training & Inference

`src/`

- preprocess.py
- pipeline.py (run_pipeline)
- evaluate.py

Entry Points

- Train: `python src/stage2_detector.py -mode train`
- Inference: `python src/pipeline.py -wsi_path slide.tif`
- Interactive: `python gradio_demo.py`

Try It Online: Gradio Demo

How to Run

```
python gradio_demo.py
```

Opens web UI at:
`http://0.0.0.0:7861`

Features

- Upload WSI patches
- View annotated results
- Download predictions
- Mitosis counting
- Nottingham score
- Browser-based
- No terminal needed

image.png

Limitations & Future Work

Current Limitations

- Domain shift: 63% F1 drop
- Single-center training
- Limited generalization

Future Directions

- Domain adaptation
- Stain augmentation
- Ensemble methods
- Multi-center trial
- Explainability (Grad-CAM)

Key Takeaways

1. Empirical Finding

Direct detection outperforms cascade
F1: 0.7742 vs 0.5806

2. Domain Generalization

63% F1 degradation on unseen stain
Multi-center validation essential

3. Best Practices

- Use Stage 2 only (F1=0.7742)
- Validate on your staining protocol
- Error analysis guides improvements
- Focal Loss + custom anchors matter

Conclusion

Thank You!

GitHub: Mitosis-Detection-TUPAC16
Demo: Gradio (http://0.0.0.0:7861)

Appendix: Evaluation Metrics

Precision, Recall, F1

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1} = 2 \cdot \frac{\text{Prec} \cdot \text{Recall}}{\text{Prec} + \text{Recall}}$$

mAP and IoU

Mean Average Precision at IoU=0.5

$\text{IoU} = \frac{\text{Overlap}}{\text{Union}}$ threshold

Appendix: Focal Loss

Standard Cross-Entropy

$$CE(p_t) = -\log(p_t)$$

Focal Loss

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$$

Key Properties

- Down-weights easy negatives (high p_t)
- Focuses training on hard cases
- Addresses class imbalance